

Week 5 Assignment

I Spy with my Little Eye

Objective

In this assignment, you will build a Python simulation of a simple espionage game between two rival intelligence agencies (**MI6** also known as SSS, and **Cheka** which evolved into KGB later). Unlike previous assignments where uncertainty came from randomness alone, this assignment introduces **strategic opponents**. One agent may deliberately mislead the other, while the opponent must use Bayesian reasoning to infer the truth.

The goal is to understand how uncertainty, deception and strategic behavior influence optimal decision making.

Mission Brief

Two rival intelligence agencies, **Agency MI6** and **Agency Cheka**, are attempting to recover a classified file. The file is hidden in one of three safehouses

$A \quad B \quad C$

Only one mission is played at a time.

Your Python program should simulate many such missions.

Simulation

For every mission, perform the following steps.

Step 1 : Hide the File

Randomly place the classified file in one of the three safehouses with equal probability.

$$P(A) = P(B) = P(C) = \frac{1}{3}.$$

Neither agency initially knows the true location.

Step 2 : MI6 Receives Intelligence

Agency MI6 receives an intelligence report.

The report is

- Correct with probability 0.6
- Incorrect with probability 0.4

If the report is incorrect, choose one of the remaining two locations uniformly at random.

For example,

Actual Location	Report Received
A	A (60%)
A	B (20%)
A	C (20%)

Step 3 : MI6 Makes Two Decisions

After receiving the report, MI6 makes two decisions

1. What message should be sent to Cheka?
2. Which safehouse should MI6 actually search?

Notice that these two decisions need not be the same.

For example,

Report	Message Sent	MI6 Searches
A	A	A
A	B	A
A	C	A
A	B	C
A	C	B

This allows MI6 to deceive Cheka while secretly searching another location.

Step 4 : Mixed Strategy

Instead of behaving deterministically, MI6 should use randomized decisions.

For example,

- Truthful message with probability

$$p_{truth} = 0.6$$

- Bluff with probability

$$1 - p_{truth} = 0.4$$

- Search according to its report with probability

$$p_{follow} = 0.7$$

- Search one of the remaining locations with probability

$$1 - p_{follow} = 0.3$$

These probabilities are parameters of MI6's strategy and can be modified later.

Step 5 : Cheka Updates its Belief

Cheka receives only the message sent by MI6.

Initially,

$$P(A) = P(B) = P(C) = \frac{1}{3}.$$

Using

- the received message,
- the probability that MI6 tells the truth,
- Bayes' Rule,

Cheka updates its belief about where the file is likely to be hidden.

Implement a function similar to

```
bayes_update(prior, message, p_truth)
```

that returns the posterior probability of every safehouse.

Step 6 : Cheka Chooses its Search

Cheka searches the safehouse having the highest posterior probability.

This represents Cheka's **best response** to the information available.

Step 7 : Reward

Award points as follows.

Outcome	MI6	Cheka
Only MI6 finds file	10	0
Only Cheka finds file	0	10
Both find file	3	3
Neither finds file	0	0

Notice that both agencies prefer to recover the file alone. Sharing the reward is less beneficial than successfully deceiving the opponent.

Step 8 : Simulation

Run the game for at least 1000 missions.

Record:-

- MI6's total score
- Cheka's total score
- Number of truthful messages
- Number of successful bluffs
- Number of failed bluffs
- Average posterior probability assigned to the true location by Cheka

Experiments

Repeat the simulation while changing the following parameters.

Experiment 1

Vary

$$p_{truth} = 0, 0.25, 0.5, 0.75, 1.$$

Observe

- MI6's score
- Cheka's score
- Bluff success rate

How does always telling the truth compare with always bluffing?

Experiment 2

Vary

$$p_{follow} = 0.5, 0.7, 0.9, 1.$$

Observe how often MI6 should trust its own intelligence report.

Does blindly following the report always lead to the highest score?

Experiment 3

Repeat the entire experiment changing the intelligence accuracy from 60% to 80% and 40%.

How does the reliability of intelligence affect the usefulness of bluffing?

Note:

In real life, Cheka may not have prior knowledge of MI6's probability of telling truth and lies. So, in that case it will have to update its belief in MI6 with each simulation.

Brief Questions

1. Explain how Bayesian updates helped Cheka make better decisions than simply trusting every message.
2. Based on your simulations, do the strategies "Always Tell the Truth" or "Always Bluff" perform well over many missions? Explain why a randomized (mixed) strategy can often be more effective.
3. Suppose, you were designing MI6's strategy. What values of p_{truth} and p_{follow} would you choose? Justify your answer using the results of your simulation.

Submission

Python source code (.py/.ipynb/.pdf). It should include answers to the Brief Questions and the inference from the three Experiments, along with Simulation results. It will not take long enough, just store your probability data in the starting and just tweak it to observe. This will teach you that only bluffing or only being honest isn't good. You have to do both. I had learned this lesson from playing cards, I hope you learn it mathematically :)